

Samina Farheen

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EDUCATION

University of California, Berkeley

M.Eng. in Mechanical Engineering — Robotics & Autonomous Systems (Fung Scholar)

Berkeley, CA

Aug. 2026 – May 2027

University of California, Santa Barbara

B.S. in Mechanical Engineering — College of Engineering Honors - GPA: 3.75

Santa Barbara, CA

Sept. 2022 – June 2026

TECHNICAL SKILLS

Modeling: SolidWorks, Siemens NX, Fusion 360, GD&T, FEA, Arena PLM, Teamcenter

Hardware: Arduino, Raspberry Pi, Sensors (encoders, pressure, accelerometers, IMUs), Actuators (DC motors, servos, solenoids)

Programming: Python, C++ (Arduino IDE), MATLAB/Simulink, ROS 2, GitHub, Linux

PROJECTS

Autonomous Fluidic Morphing Gripper Arm | *M.Eng Graduate Capstone*

Aug. 2026 – Present

- Architecting control pipelines for arm position, end-effector gripping, and fluidic morphing cell actuation into a fully autonomous system via Python-based ROS2 nodes, targeting soft object pickup with 0 deformation.
- Implementing feedforward computer vision classifier to predict 3 object stiffness profiles, targeting sub-50ms inference to drive adaptive grasp morphing during robot arm transit
- Deploying high-fidelity MuJoCo simulation models to co-design hardware-software architectures, accelerating prototyping cycles to meet a compressed 6-month delivery window.
- Leading sensor architecture with fluidic skin team by positioning high-resolution sensors to isolate elastomer-deformation noise, aiming for sub-30-second closed-loop feedback.

VIO Drone Payload (VIPER) - Sponsored by Teledyne FLIR | *Senior Capstone*

Sept. 2025 – June 2026

- Secured 1st Place at Capstone Demo and immediate sponsor R&D deployment by directing a 10-person team through a 9-month product lifecycle, synchronizing staggered ME/EE development schedules to accelerate hardware delivery
- Engineered a 286g UAV payload under a strict 300g ceiling and iPhone-scale footprint within a per-unit BOM cost of \$865 (ceiling \$1000) ensuring multi-platform modularity and meeting sponsor conditions.
- Isolated from 30-300 Hz rotor harmonics to safeguard IMU data by architecting a custom modular carbon-fiber vibration isolator in SolidWorks using water jet fabrication DFM principles, driving VIO drift error to 1.5%.
- Executed tight electronic packaging and high-density thermal management as system integrator between ME and EE teams, unlocking a continuous 45-minute flight time.

EXPERIENCE

ChargePoint, Inc.

Campbell, CA

Mechanical Design Intern

June 2025 – Sept. 2025

- Saved nearly \$2000 in labor costs by engineering an automated EV test rig data logger with Arduino electronics under a tight 4-week deadline to enable the test engineers to independently deploy test rigs
- Programmed robust C++ firmware on an Arduino Mega for 4 concurrent peripherals, configuring multi-slave SPI/I2C protocols and EEPROM storage to prevent power-outage data loss and eliminate costly test resets
- Upgraded legacy hardware via soldering and rapid prototyping of custom mounting brackets and display panel to create visual cue of cycle count for test engineers
- Designed 2 DFM/DFA-compliant sheet metal assembly jigs in Siemens NX to streamline field charger installation, reducing installation time by 15%
- Generated production-ready GD&T engineering drawings, managing via Arena PLM and NX Teamcenter, increasing hardware sprint efficiency for mechanical design team by 20%

Hawkes Lab, UC Santa Barbara

Santa Barbara, CA

Undergraduate Research Assistant

Jan. 2025 – June 2026

- Redesigned 12-unit swarm robot fleet in Fusion 360 (1 to 2 DOF) to unlock multi-directional mobility and optimized robot-to-robot interfaces to eliminate connection failure under dynamic load (0%)
- Executed force-based validation testing, processing datasets with Python and MATLAB, and co-authored empirical datasets for a graduate research publication

NASA Ames Research Center

Mountain View, CA

Rotorcraft Aeromechanics Intern

June 2022 – Aug. 2022

- Engineered a functional 1:7 scale SolidWorks assembly of an SBS test stand bed and gearbox, translating complex electromechanical systems into a tactile physical demonstrator for stakeholder presentations
- Directed and produced dynamic technical media for the HERO wildfire mitigation drone project, translating key project metrics into compelling visual demonstrations to gain funding from departmental leads